Staff answers and checks

Arrival: surface is changed, distance in centimetres is measured. Keep rover, ramp, start, release without push and measuring rule fixed. Three valid repeats are needed per condition. Keep unexpected valid results. If a run leaves the lane, preserve its note, repair the setup and repeat. A real rover faces different gravity, terrain and design.

Starter: first option, keep 51 cm with the other valid repeats. Hold the learner's first idea before explanation. Fixed data are smooth 48,51,49 cm and fabric 27,29,28 cm. They are supplied teaching evidence, not a claim that this pair measured them.

Method: low stable ramp, clear lane and catch area; same rover and marked release, no push, same measurement rule. Predict before release. Record immediately, three valid repeats per condition. Retain any failed-method note and repeat after repair. Observation or directing an adult retains the same decision and evidence demands.

Hinge: A, every valid repeat with the same measuring rule. B selects the furthest only; C changes evidence to match a prediction. Re-model if either learner is unsure. Parallel check: if one fabric value is unexpected but valid, keep it with all the others and check the method.

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Staff answers · task, limits and next move

Results desk and model: smooth mean 148/3 = 49.333… cm, fabric mean 84/3 = 28 cm. Smooth range 51−48 = 3 cm, fabric range 29−27 = 2 cm. Intervals 48–51 and 27–29 cm do not overlap. Every smooth result is larger. Lower friction opposed motion less on smooth in this model. Use actual data for an actual test and keep unexpected patterns.

Lab groups: 48,51,49 and range3cm go to Smooth; 27,29,28 and range2cm go to Fabric; smooth travelled farther and the classroom-model limitation go to Conclusion/limit. A release gate can improve consistency; it cannot by itself make one classroom rover represent every real rover.

Glitch: B, preserve every valid raw result; add a mean or range beside it. A and C hide evidence. Supported directs runs, matches six values and communicates a conclusion. Independent records three per condition and concludes with an improvement. Aspire compares means and ranges, explains variation, and evaluates transfer. All keep prediction, six results, pattern, conclusion, improvement and limit.

Exit: Supported identifies greater distance; Independent uses a repeated pattern to explain surface effect; Aspire justifies a method improvement and model limit. Hear both learners separately and retain actual feedback and pupil response. Secure: fade or transfer; mixed: contrast; misconception: counterexample and fresh check; access: change mode with same goal; method/evidence: retain note, repair and retry. Week7 reviews autumn science.

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Staff guidance · pair, access and feedback

The pair works with a shared model and individual evidence. One explains and one asks for the clue, then they swap. Hear both independent reasons before moving on. Re-model an unsure learner's step and use a fresh parallel question while the other continues.

No cold calling or forced public reading. A learner may pass. Point, sign, speak, draw, type, write, direct an adult or use exact scribing. Read the task exactly when needed. Keep the science challenge and allow planning in any language. EAL is language, not SEN.

Use a stable tray or large cards and the agreed accessible input. Give processing time. Agree a quiet supervised pause and a brief first–then cue. Re-enter at the last completed step. Do not gather evidence while a learner is distressed.

Record actual VF, WS, I, NS(+), E, R, // and ? only when they occur. Keep deepest adult prompt separate from the pupil response. A real adult receives and names back the science. R requires the pupil's response to an NS(+) next step and real Audience; never award it for a click or completion. Keep the pupil's own wording and own-colour response.

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Staff guidance · evidence and handover

Keep each first response, result or labelled explanation, the feedback and the pupil's edit or justified decision to keep it. E requires an observable curriculum action on [SYS]. Use SCI-GROW-A2-W06-P2 for written, spoken, practical or photographic evidence. Record prompt level, response mode, support faded and what was independent.

At Independent agree actual adult roles and swap across the week. If one adult attends, use the actual staffing plan. Follow individual support and behaviour plans. Record access, regulation and curriculum actions against the learner's own targets without recording a diagnosis as evidence.

If behind, preserve a prediction, six raw results, a pattern, conclusion, method improvement and model limit for each learner and the safety teaching. After a secure check shorten repeated oral rehearsal. Record any change, reason and coverage impact. Pass checked understanding, deepest prompt and unresolved errors to Week 7. Do not replace incomplete individual work with a group product.

Evaluation: date taught; delivered as planned; criteria met; actual adaptations and reasons; changes to the scheme and coverage; evidence banked. Source evidence mapping remains [AWARD] [QUAL]; no unit or qualification award is claimed by this lesson.

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